

Chapter 3

Methods

3.1 Sound Preprocessing

Traditional machine learning models are not able to handle sound data without transformation. In particular, some sort of discretization needs to be performed.

By nature, sound is continuous. Whale calls will not start perfectly at the beginning of a second and end nicely at the end of a second. Whale calls vary in duration, and some whale calls may occur simultaneously with another whale call. Figure 3.1 shows a spectrogram snippet with the corresponding manually annotated humpback whale calls, using the software Raven.

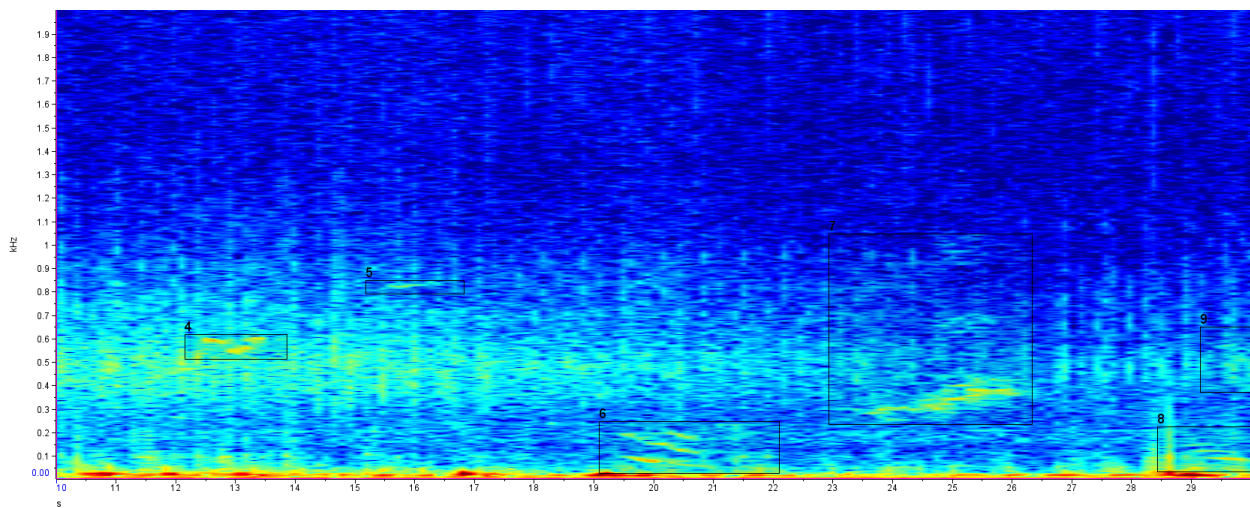


Figure 3.1 A portion of of the spectrogram visualized in Raven

One each time window, Mel frequency cepstrum coefficients are calculated. These cepstrum coefficients are calculated by first taking the discrete Fourier transform. Next, the log transformation is applied. Next, the inverse Fourier transform is applied. This gives us the cepstrum.

The cepstrum, an anagram of ‘spectrum’, is the spectrum of the log power spectrum. MFCC’s are able to emphasize

Paragraph about Mel Coefficients and the math behind them.

Earlier coefficients capture the overall spectral shape, while later coefficients capture rapid changes in sound. The figures below are scatterplots of first (Figure 3.2) and twelfth (Figure 3.3) MFCC's over the first 100 seconds.

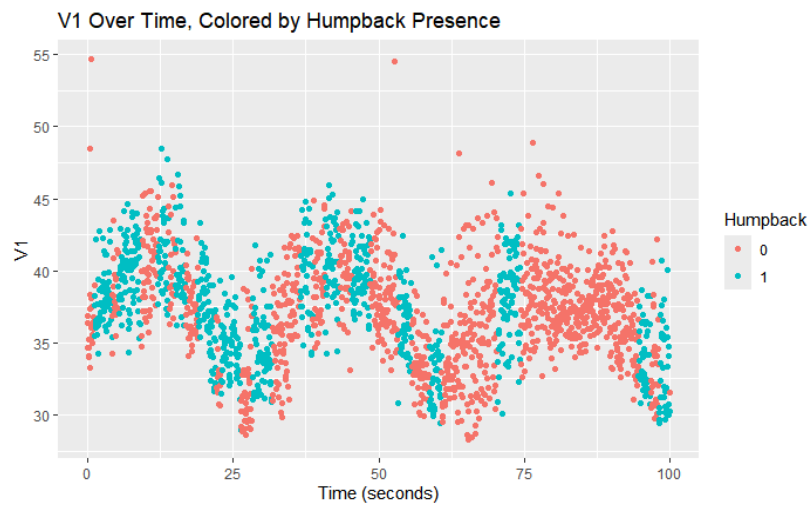


Figure 3.2: Evolution of the first MFCC over the first 100 seconds of sound

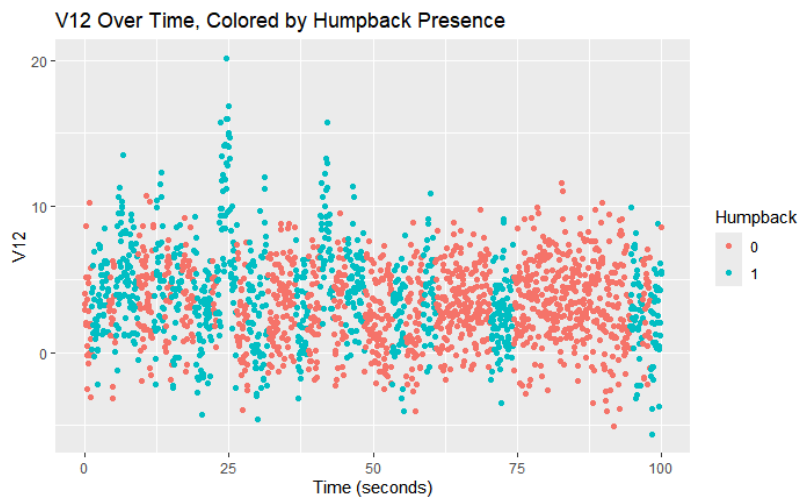


Figure 3.3: Evolution of the twelfth MFCC over the first 100 seconds of sound

The first MFCC follows a smooth structure with an apparent and predictable wave structure. The twelfth MFCC is more sporadic. Its frequency is much faster than that of the first MFCC. Large changes in value occur much more often, possibly due to new sounds being detected. Later coefficients will be helpful to identify quick changes in sound, such as whale calls. However, it may produce false positives due to detected other sounds such as other underwater creatures

3.2 Capstone Gray Whale Project

Several thirty-minute wave files containing gray whale calls were provided to the gray whale project group. These recordings occurred over the course of a week in February 2022. While higher frequency audio was captured, the observed data demonstrated that the relevant frequencies occurred at 600 Hz and below.

As mentioned previously, the capstone gray whale project split the sound data into 0.1 second chunks, then calculated MFCCs on each segment. Each segment was treated as an independent observation used to train and test machine learning models.

Their training data structure is shown below in Table 3.1:

ID	V1	V2	...	V13	Time Start	Time End	Gray Whale
1			...		0	0.1	0/1
2			...		0.1	0.2	0/1
...	...	...	...	...	...	...	...

Table 3.1: Structure of training data for the Capstone Gray Whale Project:

The thirteen Mel Coefficients are stored in the respective variables (V1 - V13). The 'Gray Whale' column is the response column, with 0's representing no whale presence, and 1's representing whale presence.

On these data, the machine learning models of random forest, k-nearest-neighbors, and an ensemble model of the two were utilized. Twenty of the thirty-minute-long files were used for training, with another ten being used for testing. These machine learning models produced their predictions for each 0.1 second interval. These individual interval predictions were not the final predictions. An additional aggregation metric was calculated, in the form of a rolling average. A rolling average of length n seconds considered all predictions in the last n seconds. Shorter rolling averages lead to more narrow and frequent prediction intervals. Longer rolling averages result in wider, less frequent prediction intervals.

For their model assessment, they took two rolling average lengths, being ‘wide’ and ‘precise’. The length of the precise intervals are ~ 8 seconds and wide intervals are ~ 30 seconds. The ultimate threshold is not explicitly stated in the paper. Both types of intervals using a random forest model based on MFCC’s resulted in a recall of 0.95. However, the wider intervals had a better precision score of 0.97, compared to the 0.90 from the precise intervals.

The end of the paper briefly mentions the challenge of humpback whales. Humpback whale calls have a larger variance in call frequencies, and other species are present during humpback whale migration. Additional or different preprocessing methods should be considered to obtain better results.

3.2.1 Gray Whale Method Applied to Humpback Whales

Before investigating additional preprocessing steps, we will investigate the results of applying the gray whale method to humpback whales.

Based on the observed data, we can safely assume almost all humpback whale songs will occur between 0 and 2000 Hz.

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3.3 Humpback Whale Additional Preprocessing

The structure used for training, testing, and evaluation did not capture enough about the nature of a humpback whale call. The machine learning models treated each row as independent, which is not ideal when a 0.1 second time segment does not capture humpback whale calls.

We build upon their method to better capture the complexity and diversity of frequencies. We start with similar preprocessing steps as the capstone project. First, we split the sound data into 0.1 second windows. However with a 0.05 second overlap. A benefit of the overlap is to minimize the amount of data lost due to unfortunate cutoff spacing. Next, we apply MFCCs to segmented sound data.

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3.3.1 Context Windows

A context window to a fixed length of data used for training (Allen et al. 2021). The goal is to effectively capture the time nature of a call. Windows too small will lead to the issue described above, in which the machine learning model could not identify a humpback whale based on only a small portion of a call. Windows too large would lead to higher odds of multiple whale calls being present in a window, which is not ideal for training purposes. Large windows will also lead to more noise surrounding a whale call, reducing the ability for the machine learning model to hone in on the whale call in the window.

For training and testing purposes context windows are considered to be positive if any whale call is present. If no humpback whale call is present within the window, then the window is treated as a negative. A good context window length provides a good intersection between effectively capturing a typical Humpback whale call, while minimizing the frequency of

capturing more than one Humpback whale call. Context windows in studies have differed in length. Factors to consider for the length of are the typical length of a whale call, how often whale calls overlap, and the overall sparsity of calls.

Context windows have been modified slightly from other papers. Other papers avoided the overlapping of context windows, and instead used a time-shift augmentation (Kather et al. 2024) to account for varying results based on the exact context window slicing.

For example, using a context window of two seconds with 12 MFCCs results in training data of the following format:

$V1_1$	$V1_2$	...	$V1_{40}$	$V2_1$	...	$V12_{40}$	Time Start	Time End	Hump-back
1							0.0	1.0	
2							0.5	1.5	
3							1.0	2.0	
							...	...	
40							1.9	2.0	

Table 3.2: Table used for Training Data

Each row captures the evolutions of a MFCCs over the context window.

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We train machine learning models on these context windows.

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3.4 Model Fitting

Models were tested using linear discriminant analysis, k-nearest-neighbors, and random forest.

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3.5 Model Assessing and Aggregation

While the initial results are helpful, additional aggregation of results are performed. We use the same rolling evaluation windows as used in (Wong et al. 2025).

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